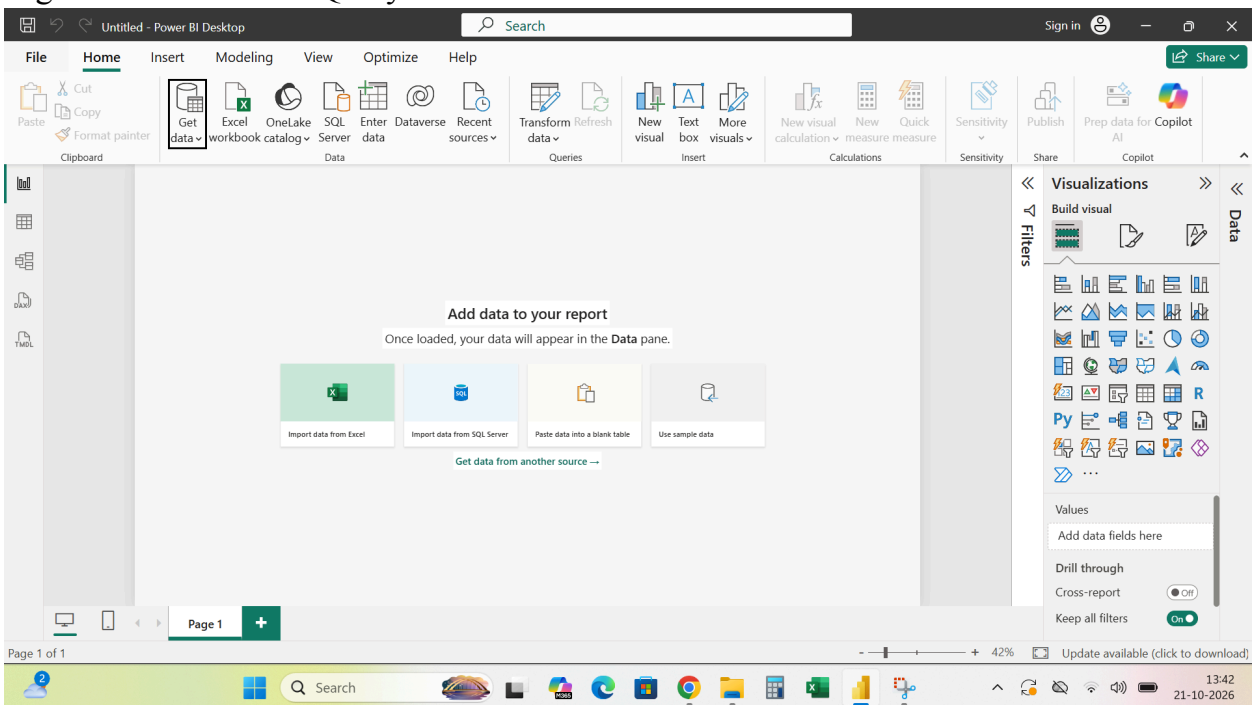
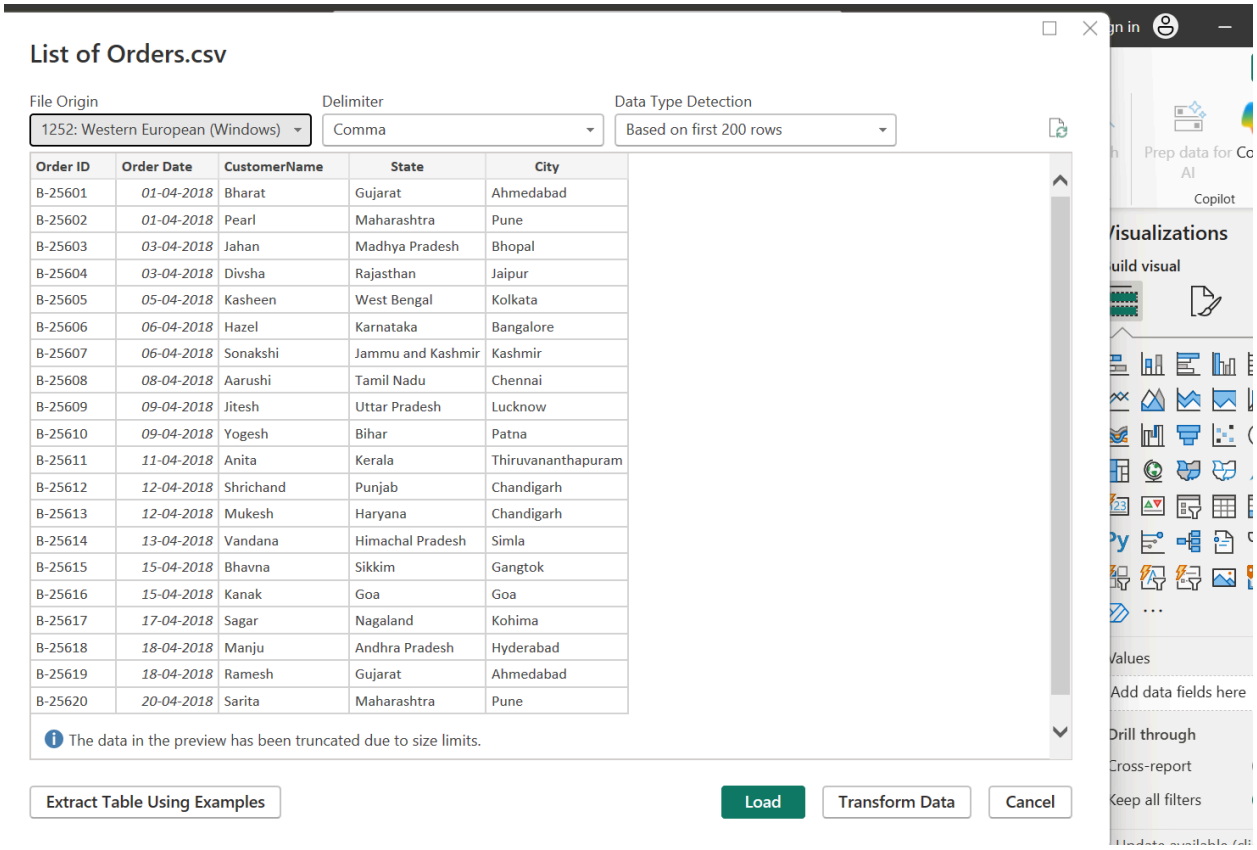
Power BI Assignment 1 -Data Transformation & Data Modeling

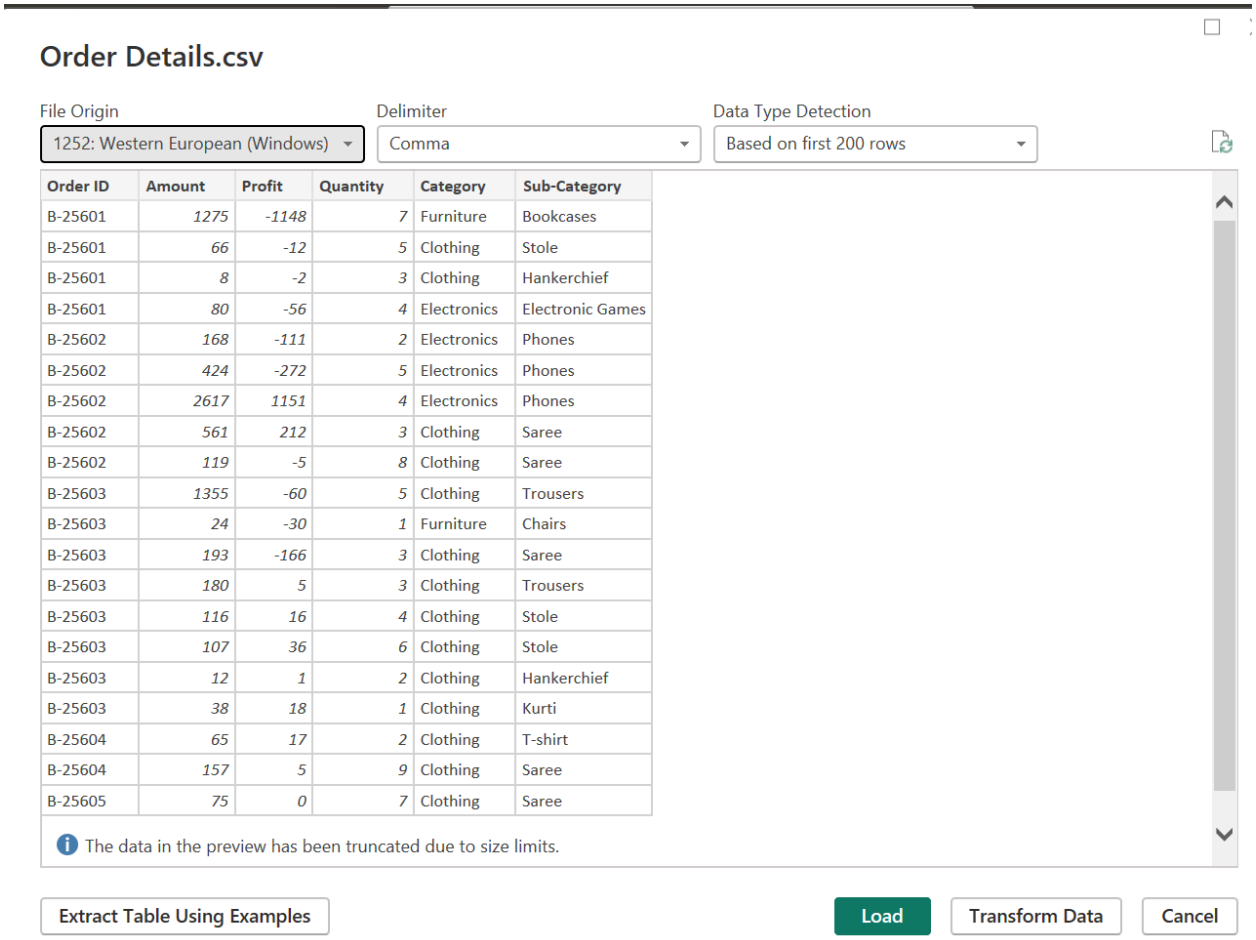
Data Transformation:

1.Click the Power Bi and click the Get data ,Imported the “Order Details.csv” and “Sales target.csv” into Power Query Editor.





Power BI Assignment 1 -Data Transformation & Data Modeling



Order Details.csv

File Origin: 1252: Western European (Windows)

Delimiter: Comma

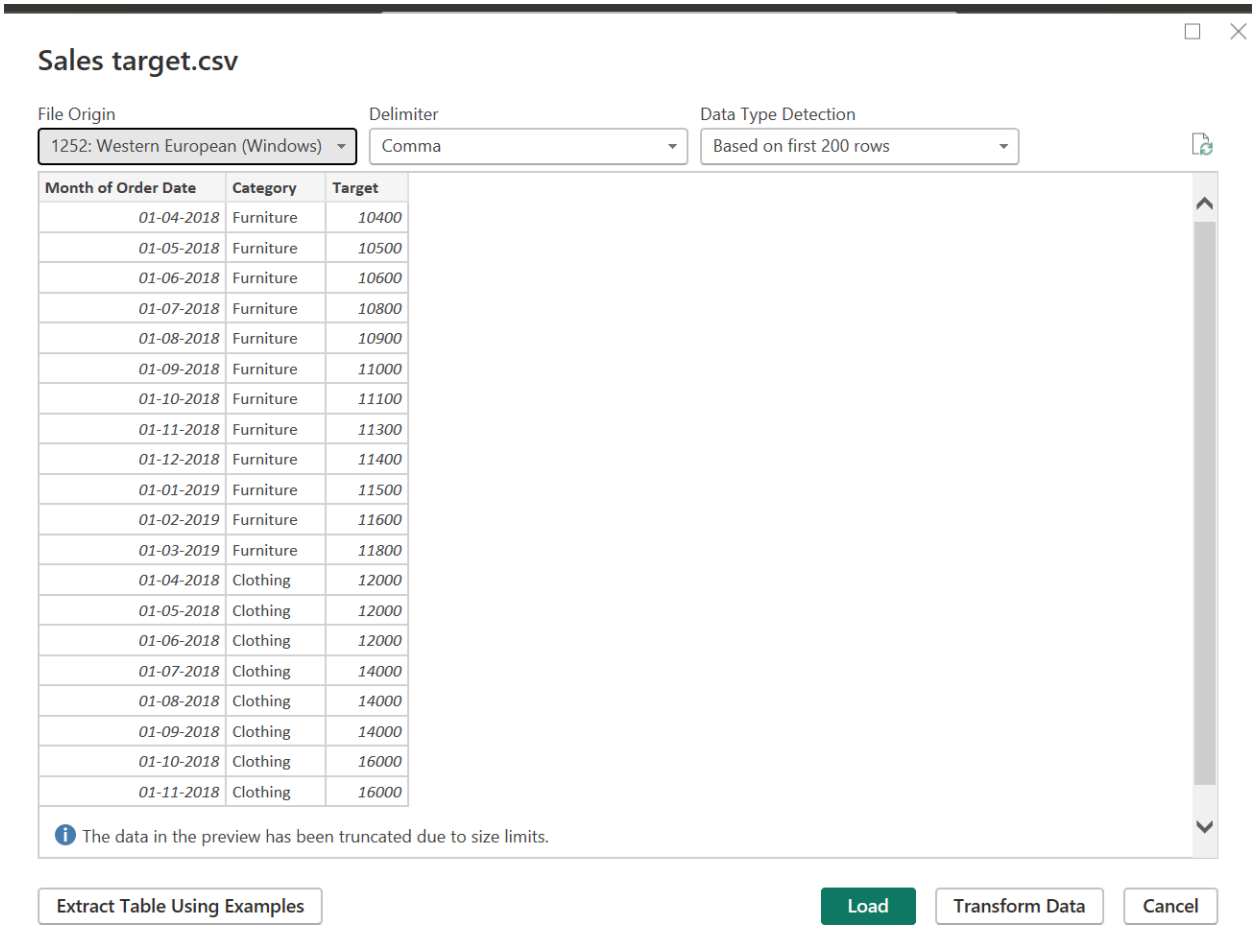
Data Type Detection: Based on first 200 rows

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic Games
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

Power BI Assignment 1 -Data Transformation & Data Modeling



Sales target.csv

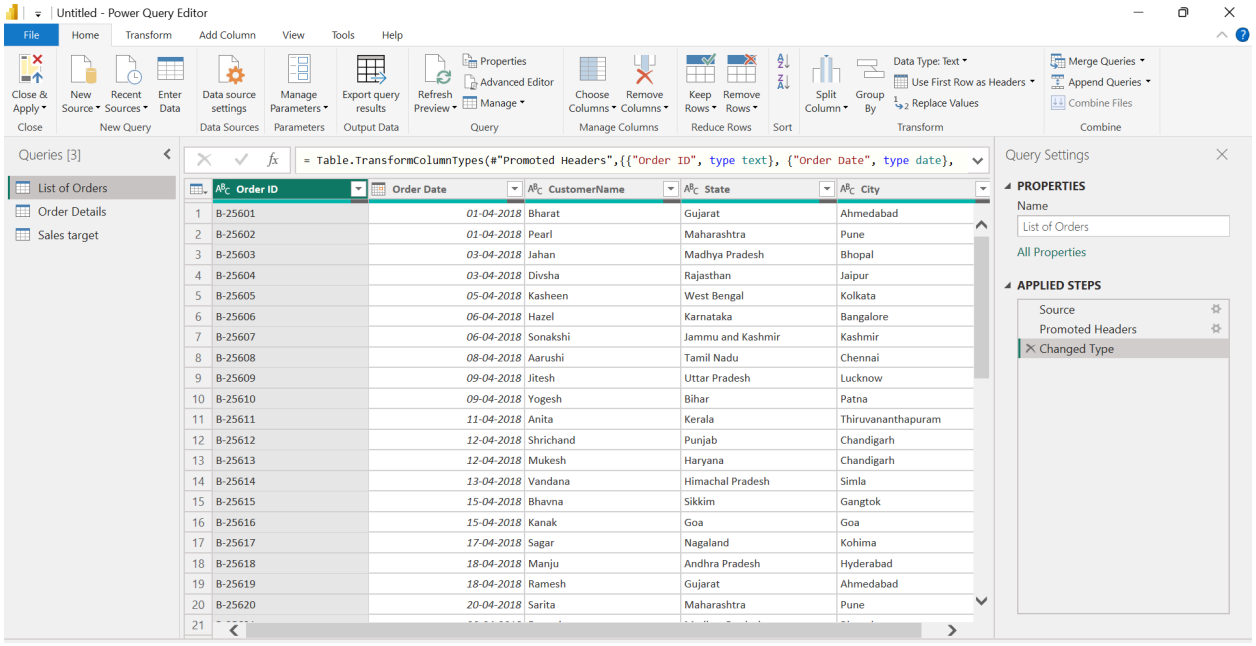
File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Month of Order Date	Category	Target
01-04-2018	Furniture	10400
01-05-2018	Furniture	10500
01-06-2018	Furniture	10600
01-07-2018	Furniture	10800
01-08-2018	Furniture	10900
01-09-2018	Furniture	11000
01-10-2018	Furniture	11100
01-11-2018	Furniture	11300
01-12-2018	Furniture	11400
01-01-2019	Furniture	11500
01-02-2019	Furniture	11600
01-03-2019	Furniture	11800
01-04-2018	Clothing	12000
01-05-2018	Clothing	12000
01-06-2018	Clothing	12000
01-07-2018	Clothing	14000
01-08-2018	Clothing	14000
01-09-2018	Clothing	14000
01-10-2018	Clothing	16000
01-11-2018	Clothing	16000

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

2.Transform the data



Untitled - Power Query Editor

Queries [3]: List of Orders, Order Details, Sales target

```
= Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Order Date", type date},
```

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21					

Query Settings

PROPERTIES

Name: List of Orders

All Properties

APPLIED STEPS

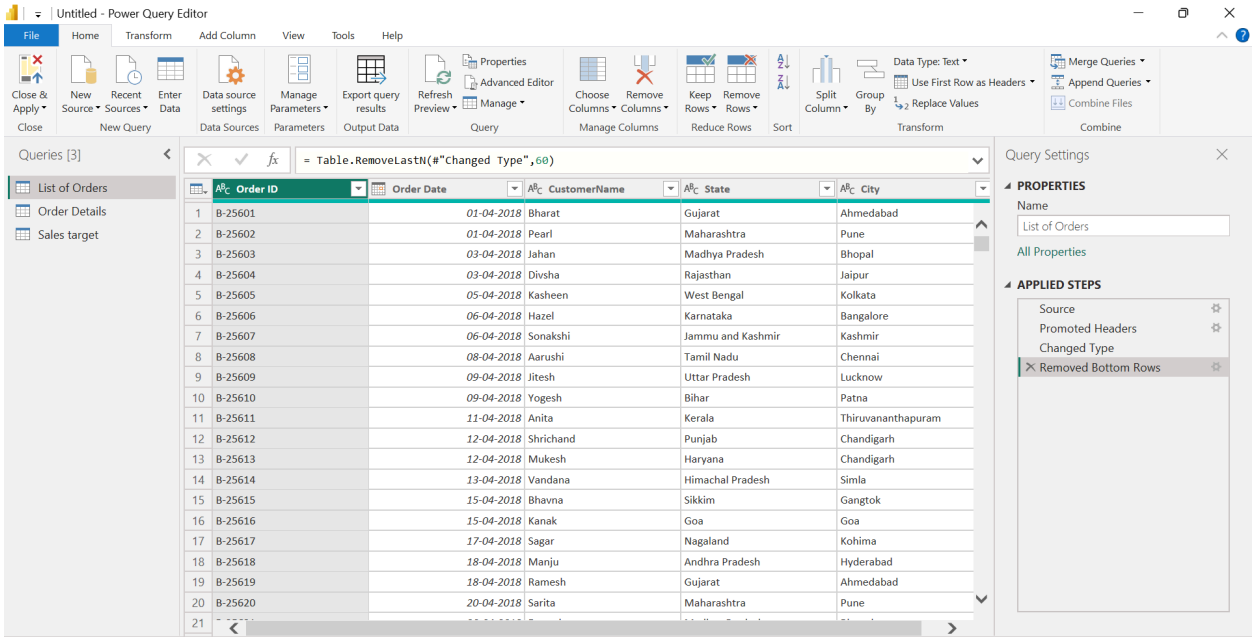
Source

Promoted Headers

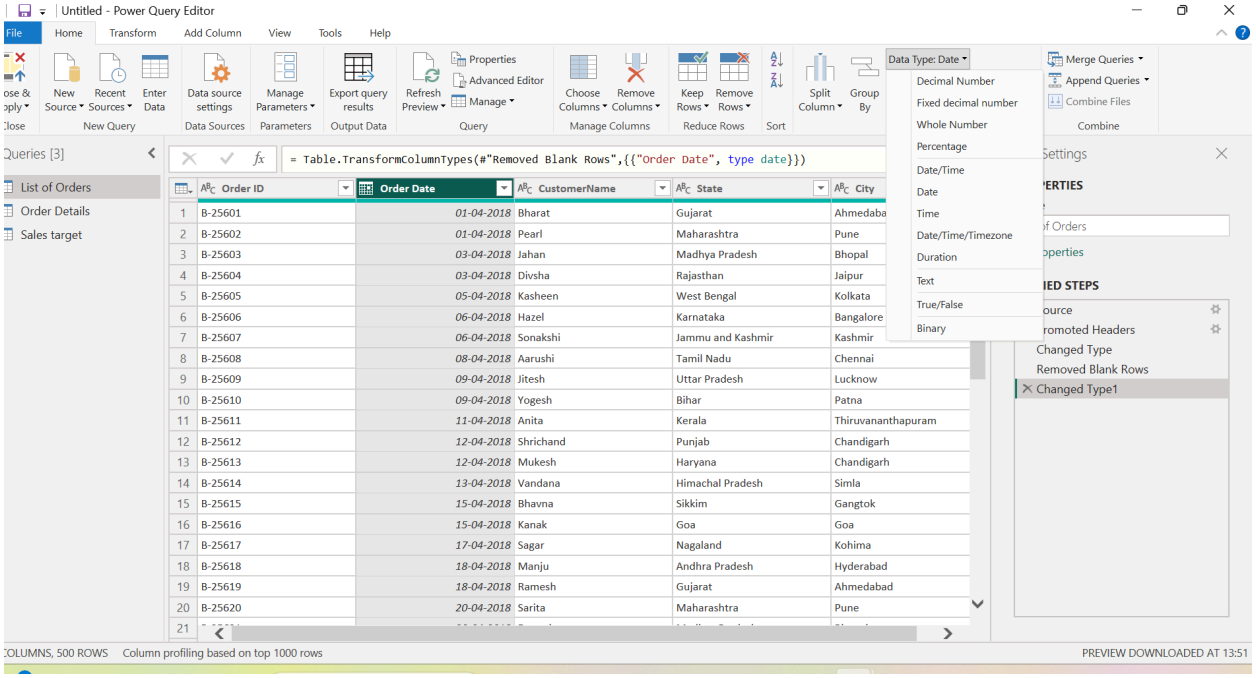
Changed Type

Power BI Assignment 1 -Data Transformation & Data Modeling

3.Start to the cleaning data Removed the bottom rows

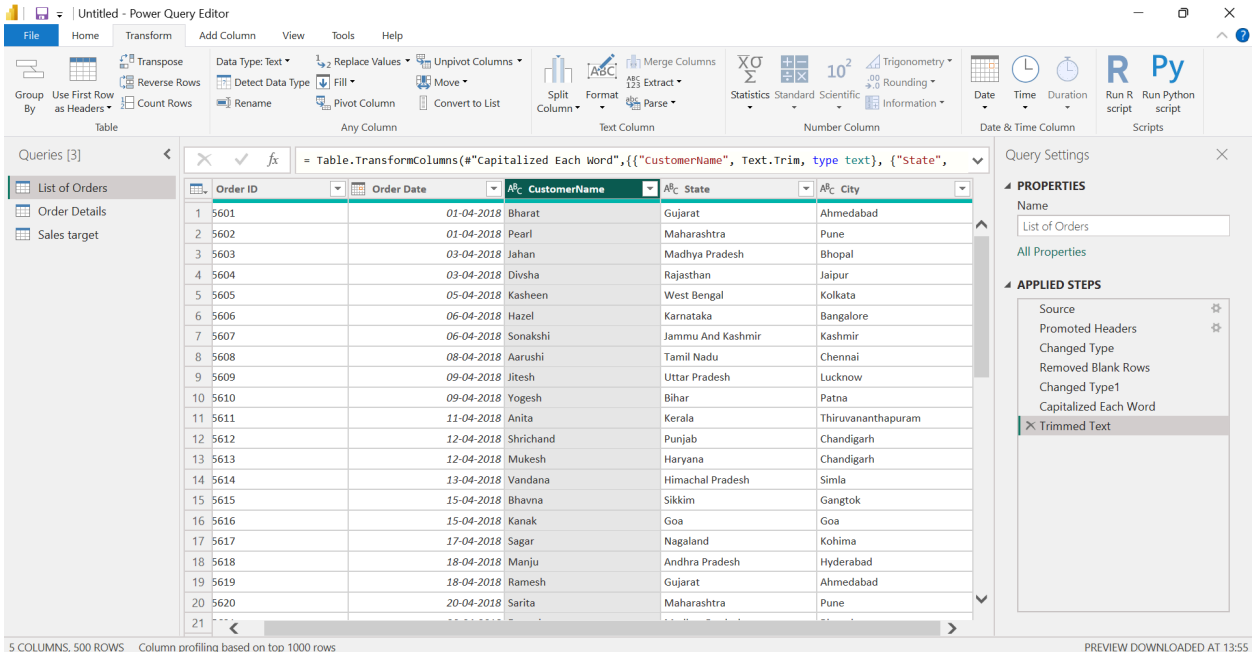


4.Changed the data type to date

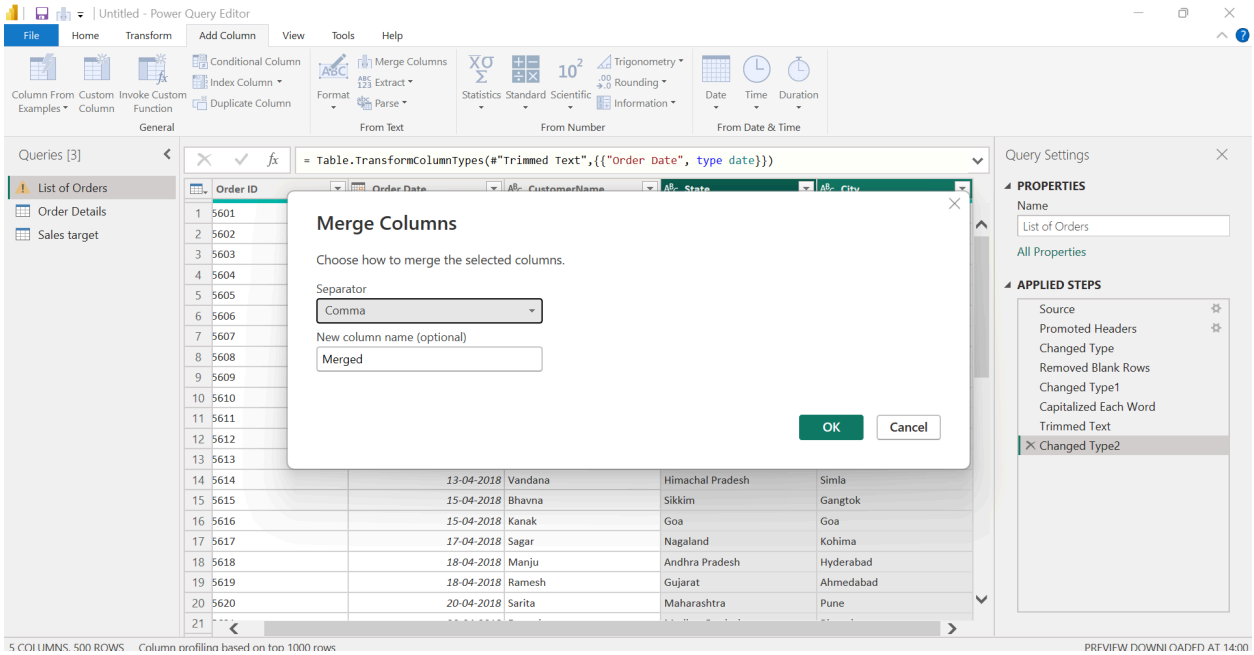


5.Changed the "CustomerName" column into proper case

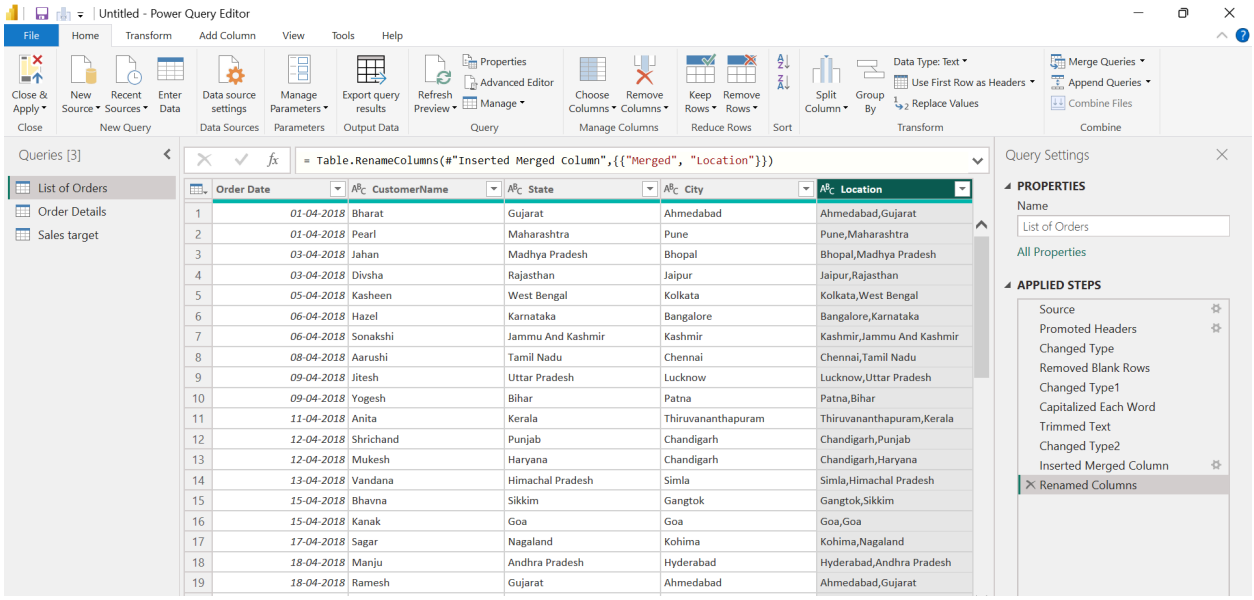
Power BI Assignment 1 -Data Transformation & Data Modeling



6.Merged the "State" and "City" columns to create a new column named "Location"

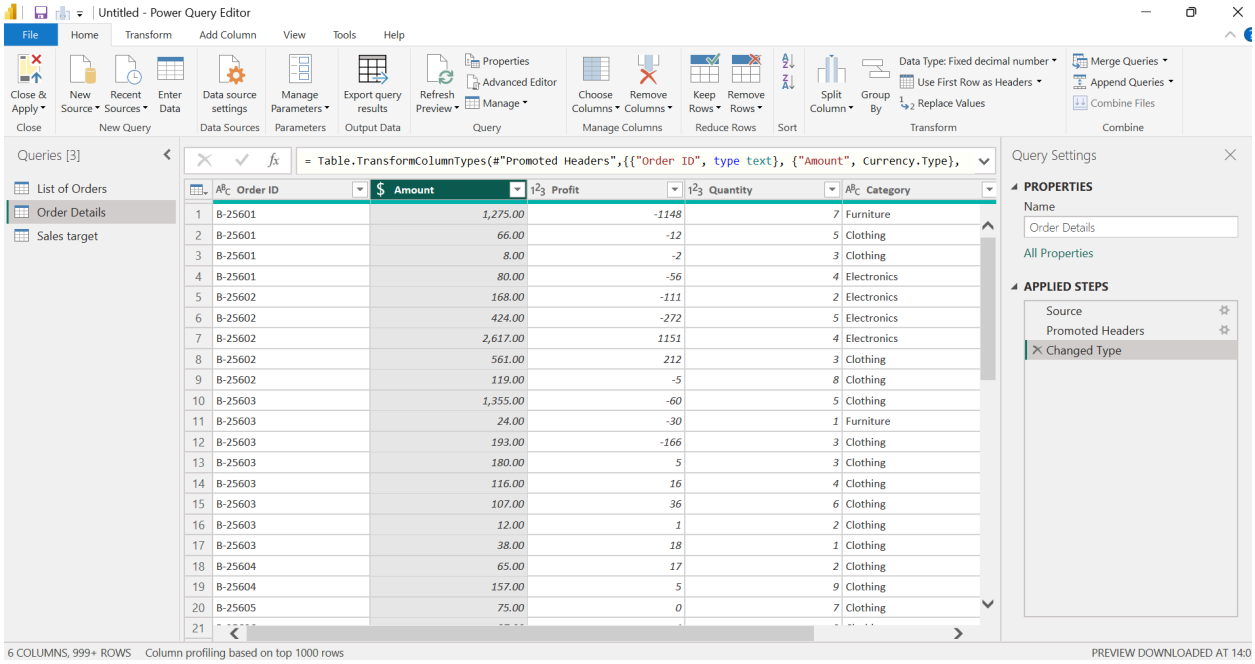


Power BI Assignment 1 -Data Transformation & Data Modeling



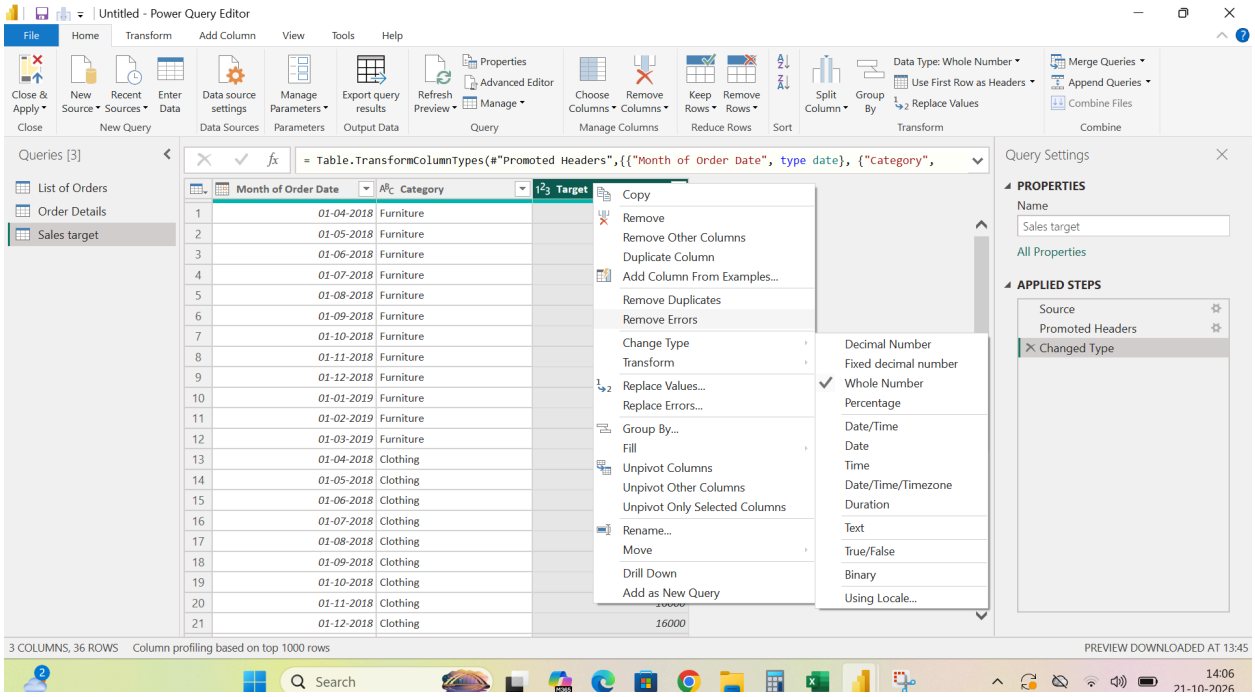
	Order Date	CustomerName	State	City	Location
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7	06-04-2018	Sonakshi	Jammu And Kashmir	Kashmir	Kashmir,Jammu And Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat

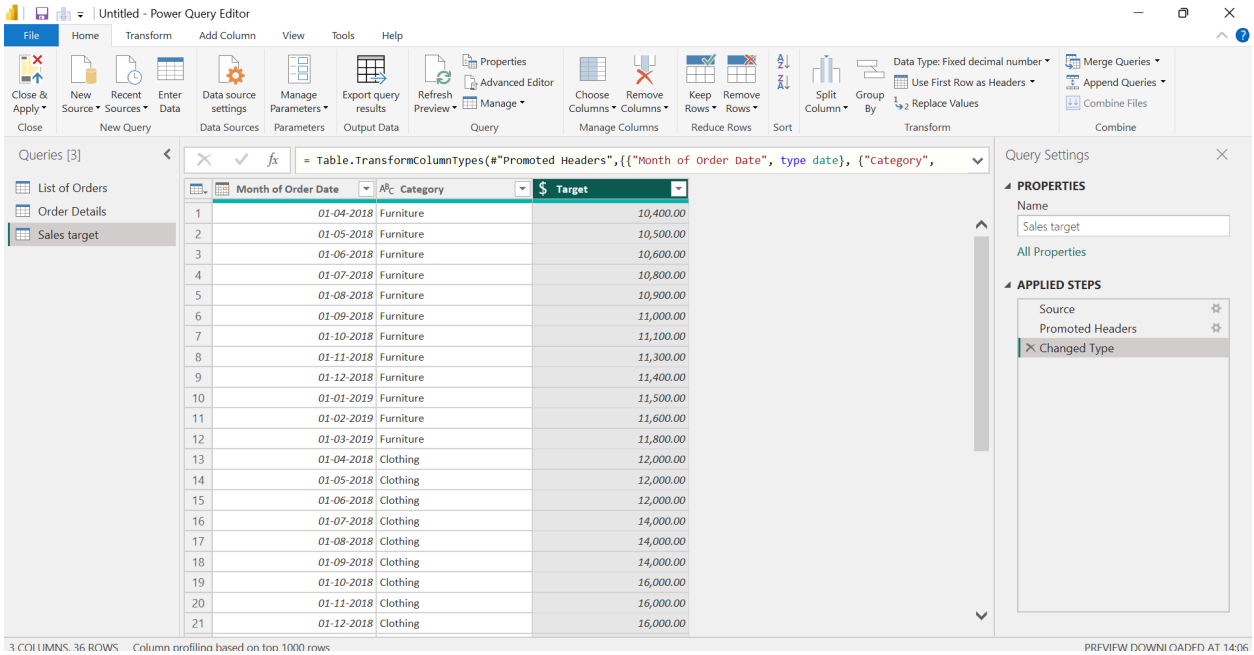
7.Changed the data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.



	Order ID	Amount	Profit	Quantity	Category
1	B-25601	1,275.00	-1148	7	Furniture
2	B-25601	66.00	-12	5	Clothing
3	B-25601	8.00	-2	3	Clothing
4	B-25601	80.00	-56	4	Electronics
5	B-25602	168.00	-111	2	Electronics
6	B-25602	424.00	-272	5	Electronics
7	B-25602	2,617.00	1151	4	Electronics
8	B-25602	561.00	212	3	Clothing
9	B-25602	119.00	-5	8	Clothing
10	B-25603	1,355.00	-60	5	Clothing
11	B-25603	24.00	-30	1	Furniture
12	B-25603	193.00	-166	3	Clothing
13	B-25603	180.00	5	3	Clothing
14	B-25603	116.00	16	4	Clothing
15	B-25603	107.00	36	6	Clothing
16	B-25603	12.00	1	2	Clothing
17	B-25603	38.00	18	1	Clothing
18	B-25604	65.00	17	2	Clothing
19	B-25604	157.00	5	9	Clothing
20	B-25605	75.00	0	7	Clothing

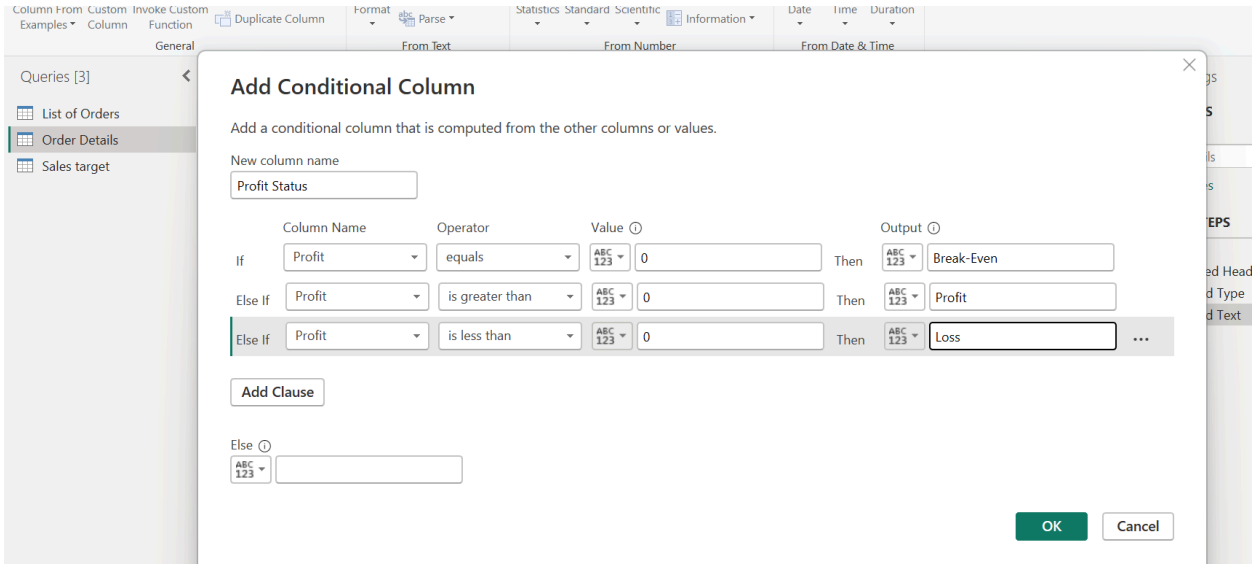
Power BI Assignment 1 -Data Transformation & Data Modeling



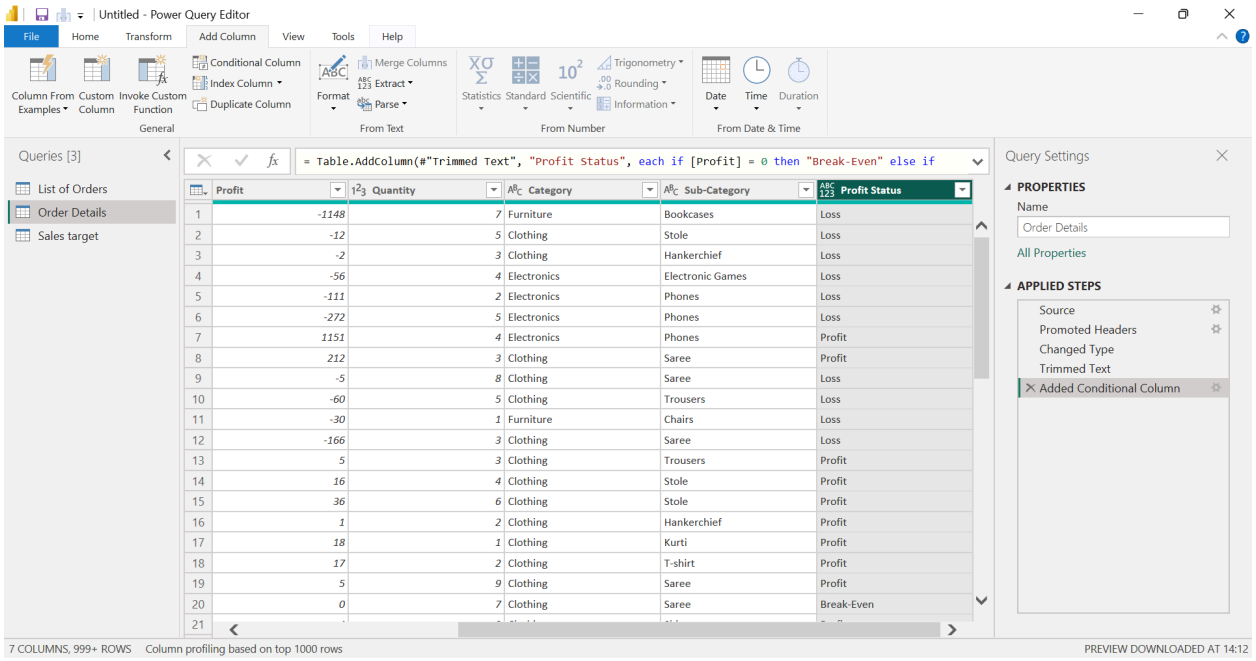


8.Added a new conditional column named "Profit Status" based on the values in the "Profit" column. Change the type :if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

Power BI Assignment 1 -Data Transformation & Data Modeling



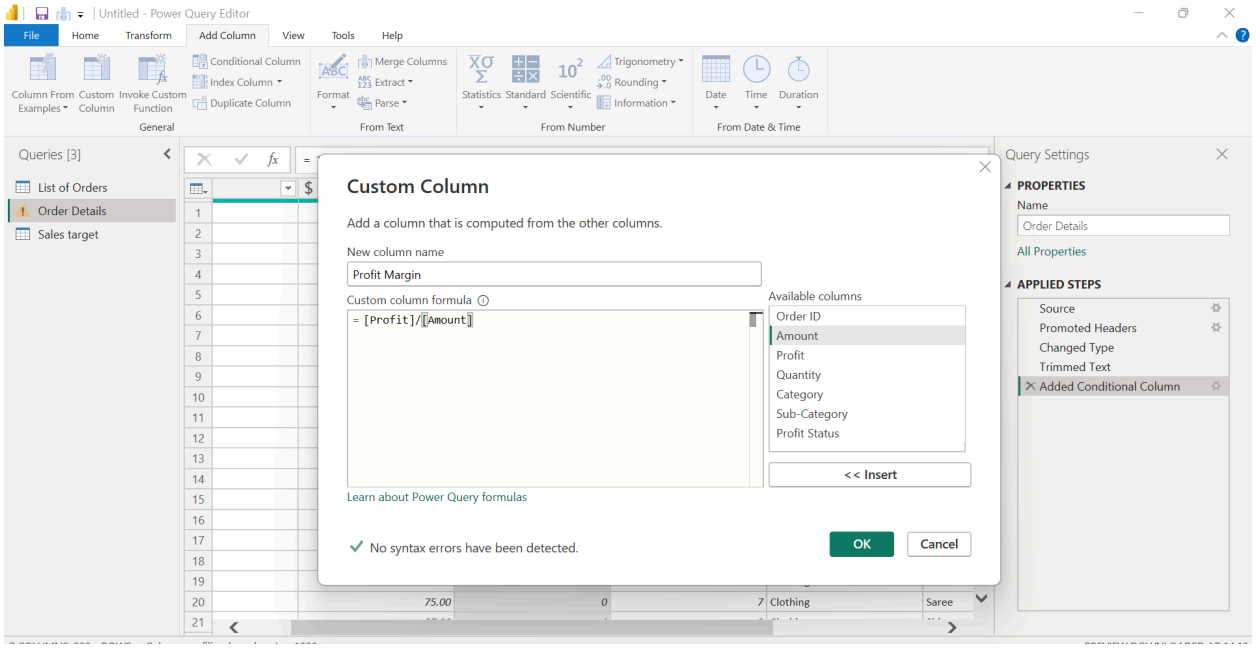
After that a new column was created.



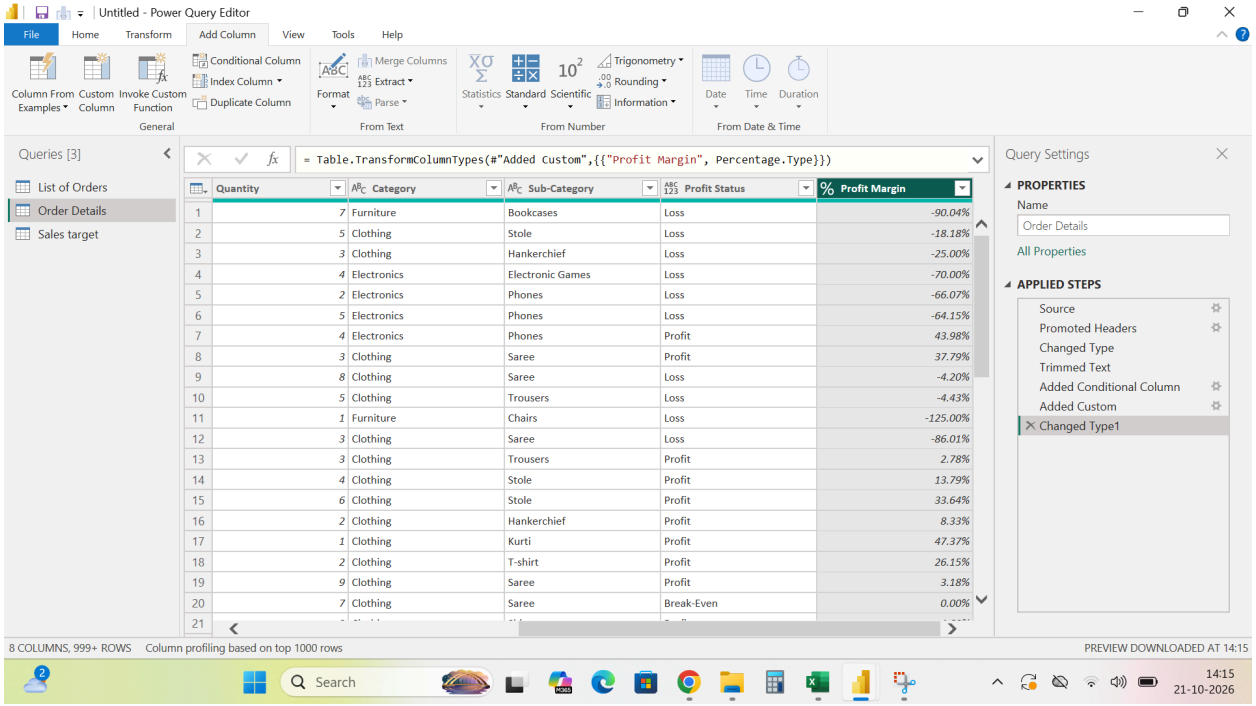
	Profit	Quantity	Category	Sub-Category	Profit Status
1	-1148	7	Furniture	Bookcases	Loss
2	-12	5	Clothing	Stole	Loss
3	-2	3	Clothing	Hankerchief	Loss
4	-56	4	Electronics	Electronic Games	Loss
5	-111	2	Electronics	Phones	Loss
6	-272	5	Electronics	Phones	Loss
7	1151	4	Electronics	Phones	Profit
8	212	3	Clothing	Saree	Profit
9	-5	8	Clothing	Saree	Loss
10	-60	5	Clothing	Trousers	Loss
11	-30	1	Furniture	Chairs	Loss
12	-166	3	Clothing	Saree	Loss
13	5	3	Clothing	Trousers	Profit
14	16	4	Clothing	Stole	Profit
15	36	6	Clothing	Stole	Profit
16	1	2	Clothing	Hankerchief	Profit
17	18	1	Clothing	Kurti	Profit
18	17	2	Clothing	T-shirt	Profit
19	5	9	Clothing	Saree	Profit
20	0	7	Clothing	Saree	Break-Even
21					

9.Created a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".

Power BI Assignment 1 -Data Transformation & Data Modeling



10.After that changed the percentage

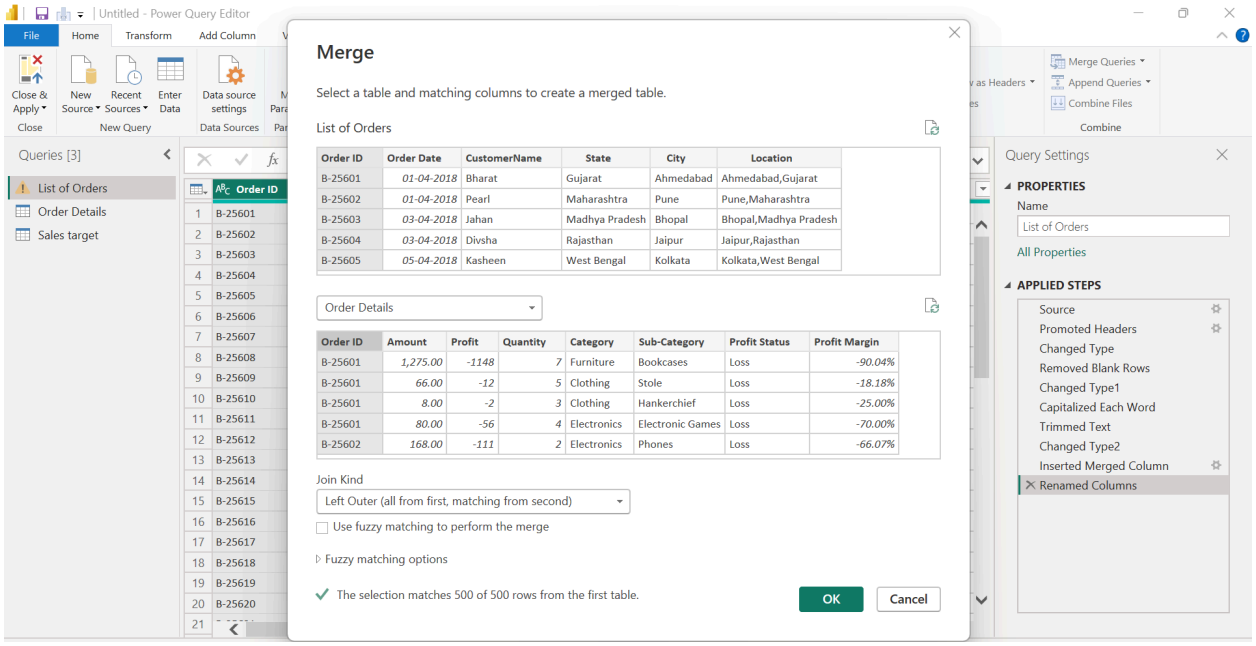


	Quantity	Category	Sub-Category	Profit Status	Profit Margin
1	7	Furniture	Bookcases	Loss	-90.04%
2	5	Clothing	Stole	Loss	-18.18%
3	3	Clothing	Hankerchief	Loss	-25.00%
4	4	Electronics	Electronic Games	Loss	-70.00%
5	2	Electronics	Phones	Loss	-66.07%
6	5	Electronics	Phones	Loss	-64.15%
7	4	Electronics	Phones	Profit	43.98%
8	3	Clothing	Saree	Profit	37.79%
9	8	Clothing	Saree	Loss	-4.20%
10	5	Clothing	Trousers	Loss	-4.43%
11	1	Furniture	Chairs	Loss	-125.00%
12	3	Clothing	Saree	Loss	-86.01%
13	3	Clothing	Trousers	Profit	2.78%
14	4	Clothing	Stole	Profit	13.79%
15	6	Clothing	Stole	Profit	33.64%
16	2	Clothing	Hankerchief	Profit	8.33%
17	1	Clothing	Kurti	Profit	47.37%
18	2	Clothing	T-shirt	Profit	26.15%
19	9	Clothing	Saree	Profit	3.18%
20	7	Clothing	Saree	Break-Even	0.00%

Merging Data :

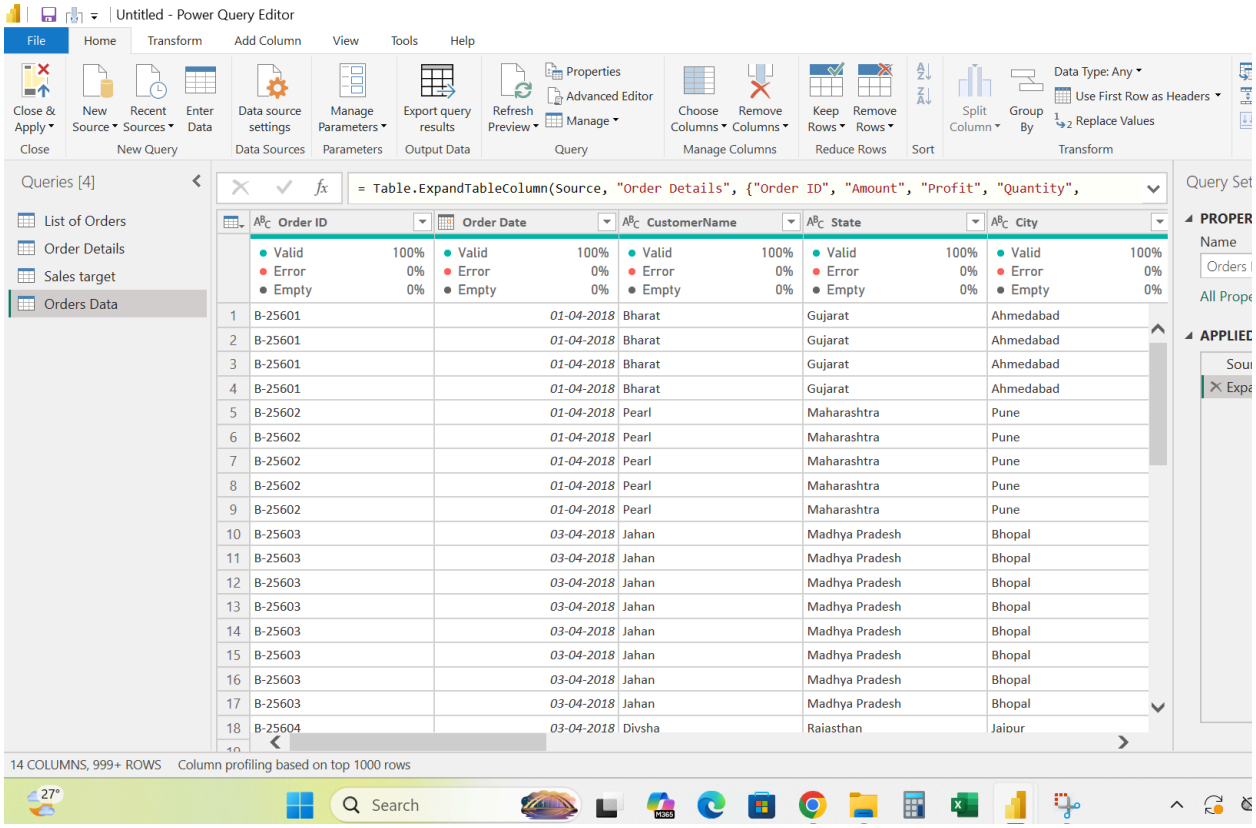
11.Merged the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

Power BI Assignment 1 -Data Transformation & Data Modeling

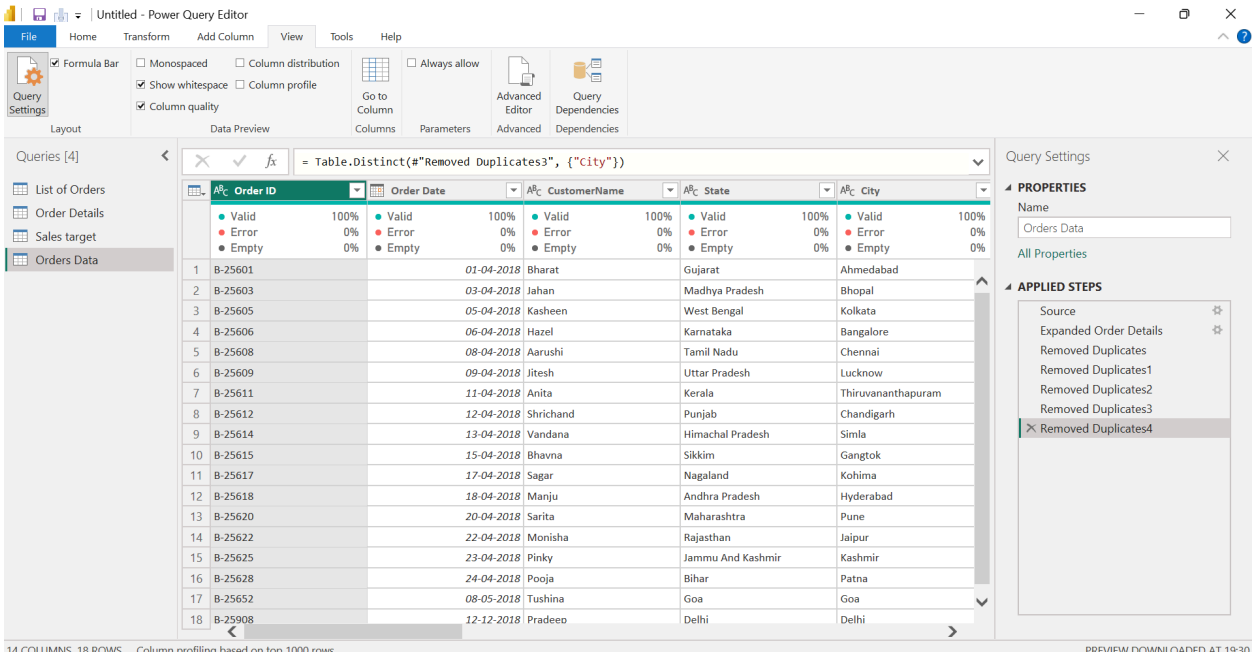


Handling Missing Data & Duplicate Data:

12. No null values



Power BI Assignment 1 -Data Transformation & Data Modeling

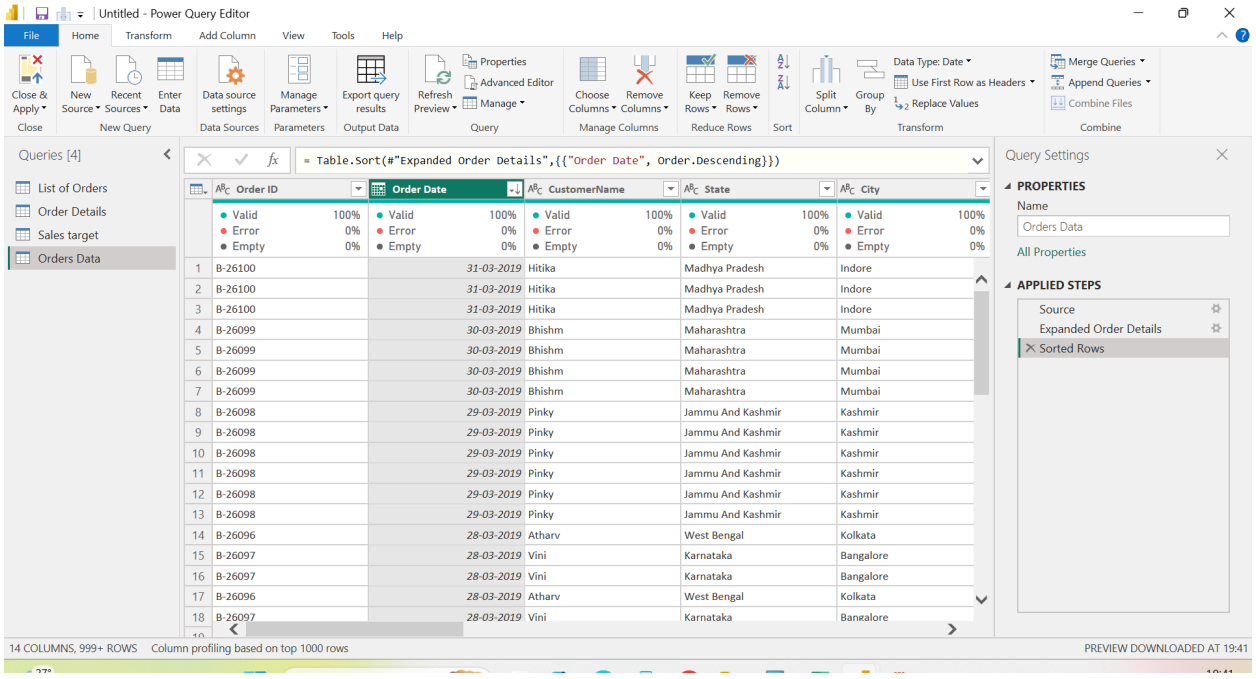


	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
3	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
4	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
5	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
6	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
7	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
8	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
9	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
10	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
11	B-25617	17-04-2018	Sagar	Nagaland	Kohima
12	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
13	B-25620	20-04-2018	Sarita	Maharashtra	Pune
14	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
15	B-25625	23-04-2018	Pinky	Jammu And Kashmir	Kashmir
16	B-25628	24-04-2018	Pooja	Bihar	Patna
17	B-25652	08-05-2018	Tushina	Goa	Goa
18	B-25908	12-12-2018	Pradeep	Delhi	Delhi

No duplicates Data found

Sorting and Filtering Data:

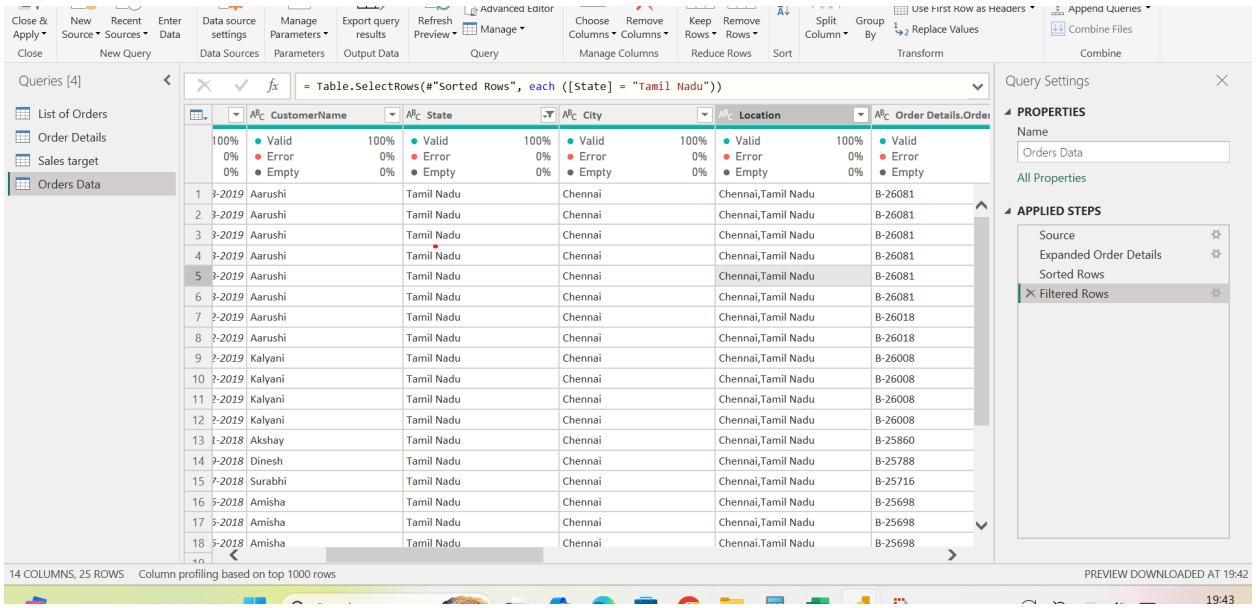
Sort the orders by Order Date in descending order



	Order ID	Order Date	CustomerName	State	City
1	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore
2	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore
3	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore
4	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
5	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
6	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
7	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
8	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
9	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
10	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
11	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
12	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
13	B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir
14	B-26096	28-03-2019	Atharv	West Bengal	Kolkata
15	B-26097	28-03-2019	Vini	Karnataka	Bangalore
16	B-26097	28-03-2019	Vini	Karnataka	Bangalore
17	B-26096	28-03-2019	Atharv	West Bengal	Kolkata
18	B-26097	28-03-2019	Vini	Karnataka	Bangalore

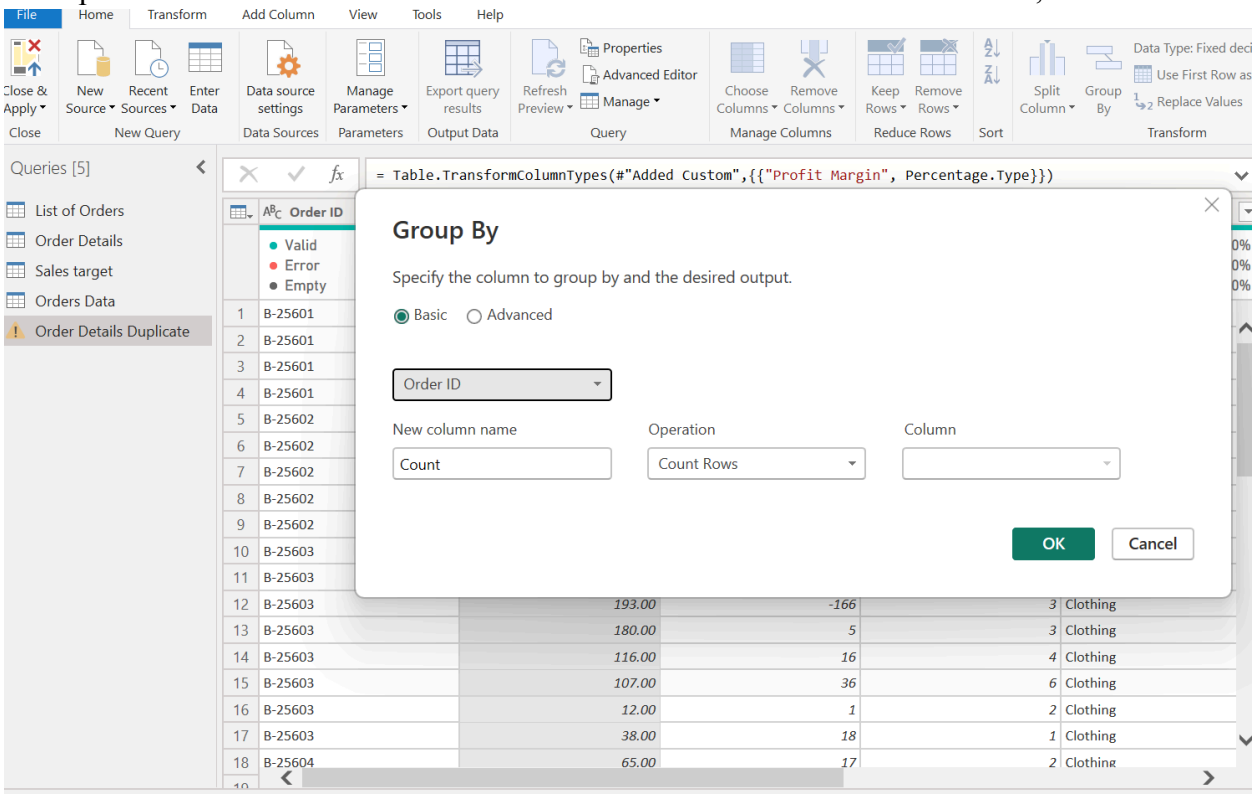
13.Filter the orders to focus only on a specific state Tamil Nadu for regional analysis.

Power BI Assignment 1 -Data Transformation & Data Modeling

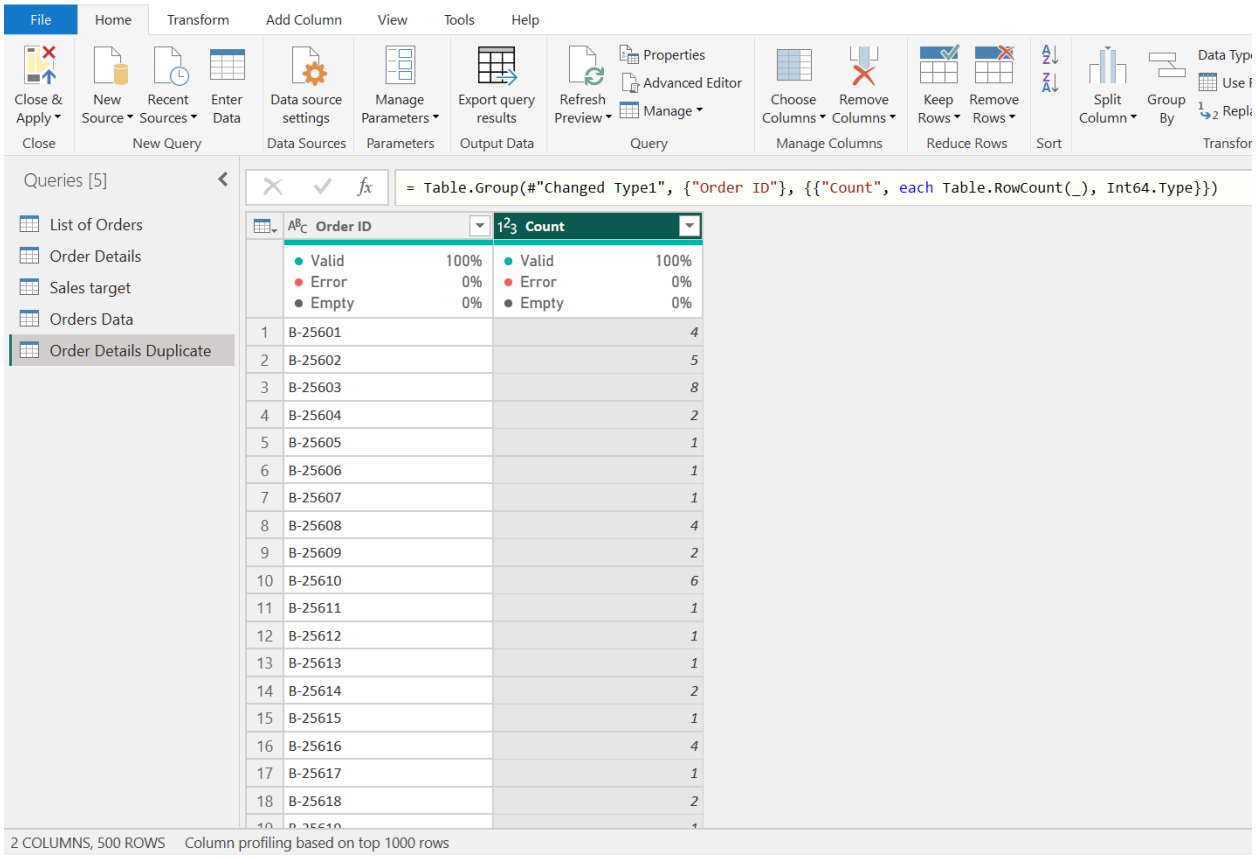


Grouping and Aggregating Data:

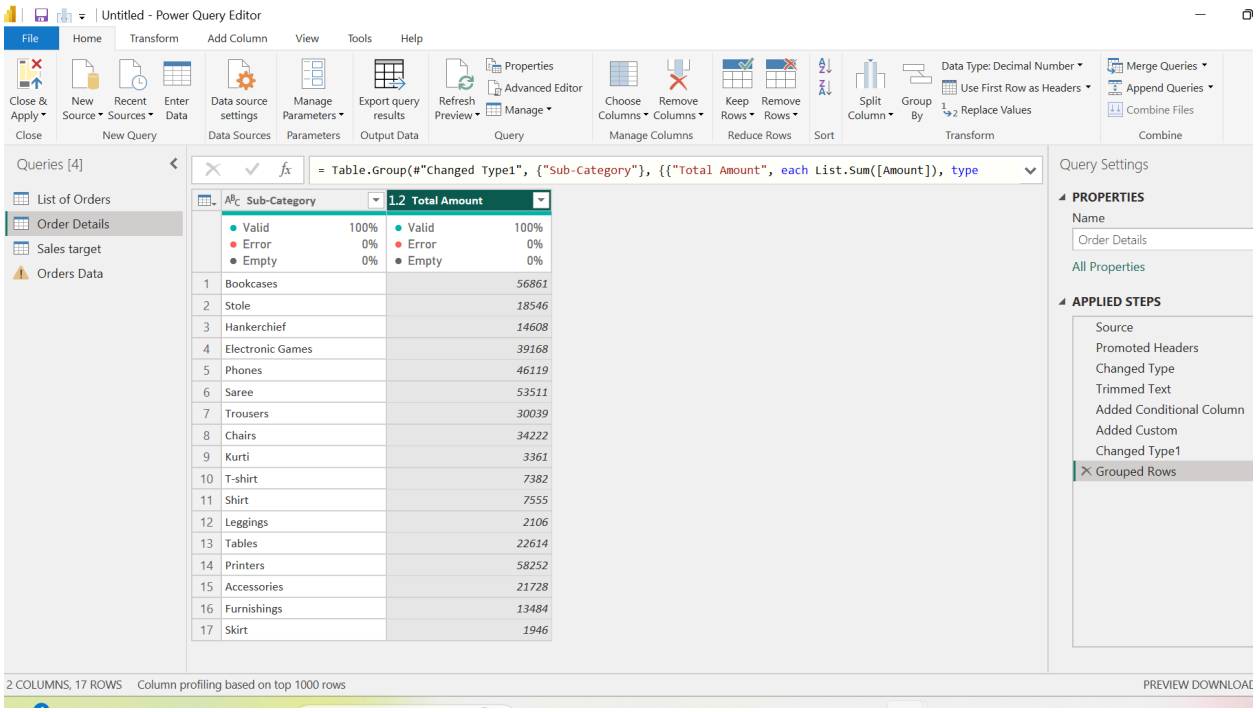
14.Duplicate the "Order Details" table and calculated the count of each Order ID,



Power BI Assignment 1 -Data Transformation & Data Modeling

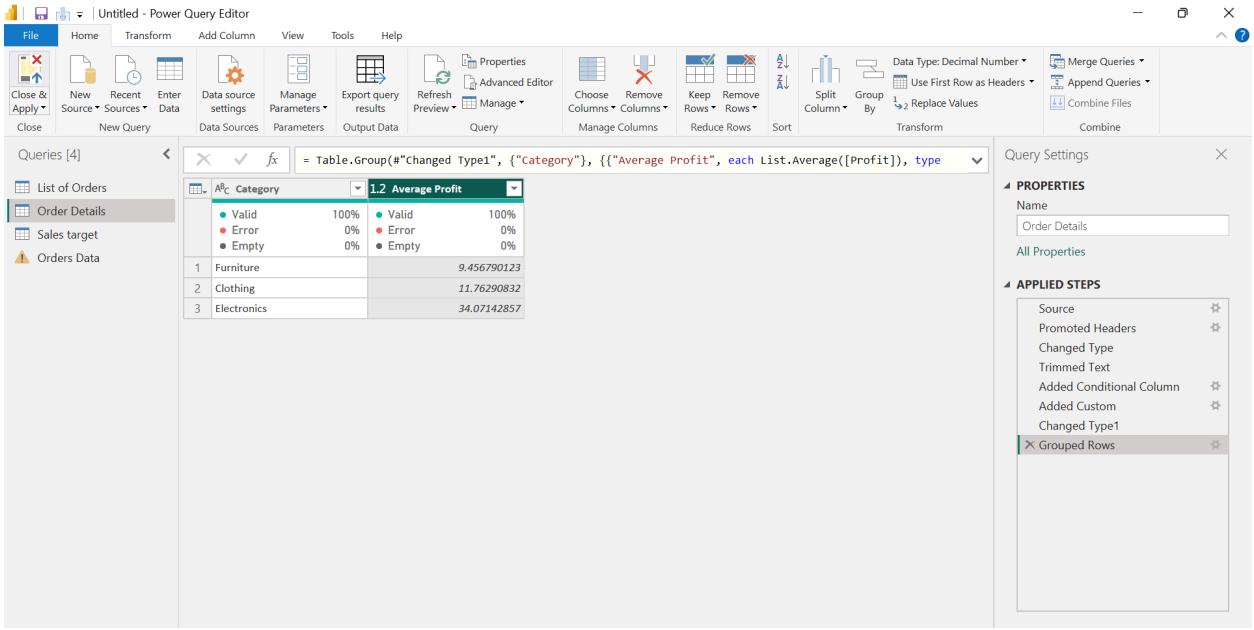


15. total amount by Sub-Category.

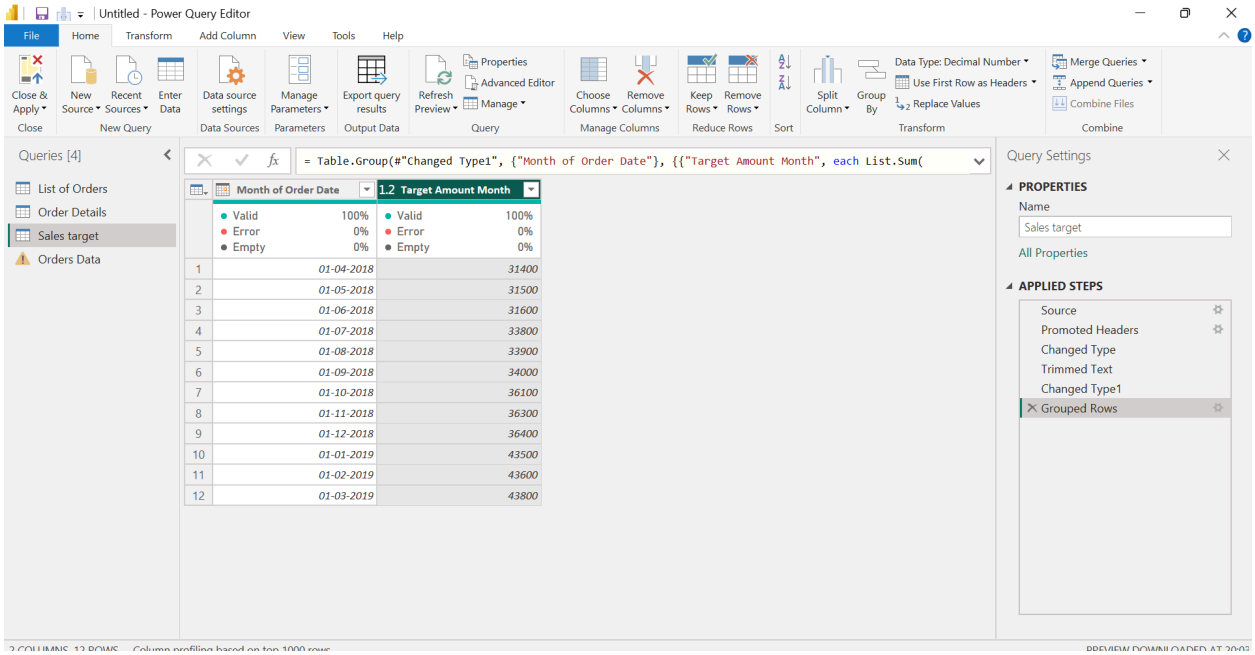


16.average profit by Category

Power BI Assignment 1 -Data Transformation & Data Modeling



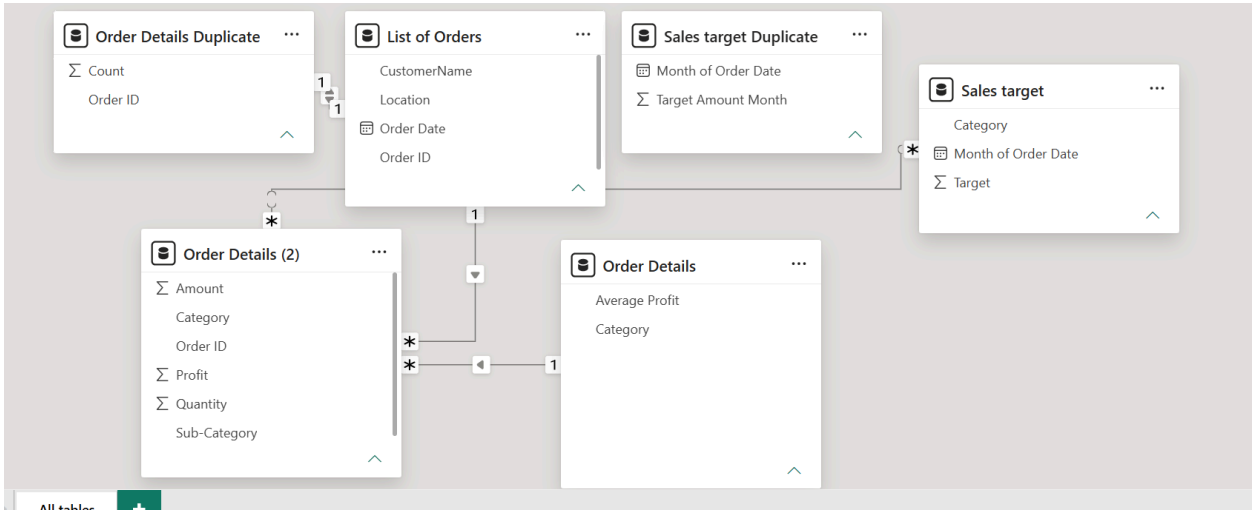
15.Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.



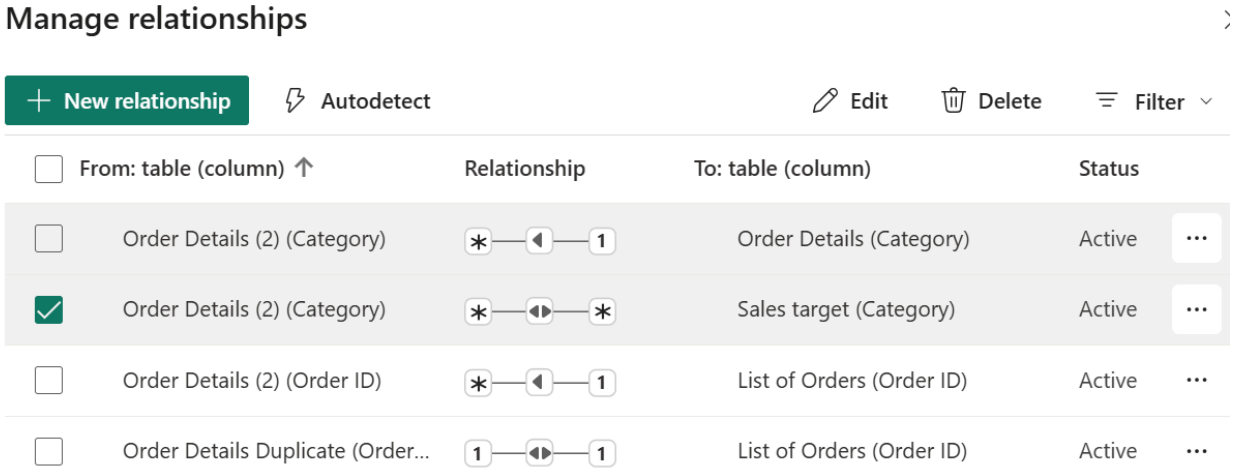
Data Modeling:

Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

Power BI Assignment 1 -Data Transformation & Data Modeling



Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.



Power BI Assignment 1 -Data Transformation & Data Modeling

